MARKING SCHEME — BCSE/22-1/MOCK

Class X Physics · Mock Paper (for self-checking)

Section A — Q1(a) Multiple choice (1 mark each)

No.	Answer	Reason
i.	**B**	Moment = force × distance, so the unit is N m.
ii.	**A**	Stable equilibrium: C.G. rises and the body returns.
iii.	**D**	$6 \times 40 = W \times 20 \Rightarrow W = 12$ N.
iv.	**B**	Low C.G. and a wide base give the greatest stability.
v.	**C**	$F \propto 1/r^2$: $(1/\frac{1}{2})^2 = 4$.
vi.	**C**	$v = g\,t = 10 \times 3 = 30$ m/s.
vii.	**B**	Terminal velocity: drag = weight, net force zero.
viii.	**D**	$W = F\,d = 50 \times 4 = 200$ J.
ix.	**B**	$W = 500 \times 10 \times 10 = 50\,000$ J; $P = 50\,000 / 20 = 2500$ W.
x.	**B**	$400 / 500 \times 100 = 80$ %.
xi.	**C**	$P = F/A = 200 / 0.5 = 400$ Pa.
xii.	**D**	$F_2 = 20 \times 40/2 = 400$ N.
xiii.	**B**	$P = \rho\,g\,h$.
xiv.	**C**	$Q = 3 \times 4200 \times 10 = 126\,000$ J.
xv.	**D**	$\Delta T = Q/mc$: the largest c gives the smallest rise.
xvi.	**C**	Latent heat of fusion: temperature stays constant.
xvii.	**A**	Resistance = V/I = slope; P is the steepest.
xviii.	**B**	$I = V/R = 36/12 = 3$ A.
xix.	**C**	$H = I^2Rt = 4 \times 5 \times 10 = 200$ J.
xx.	**B**	Left hand = motor effect (force).
xxi.	**C**	$V_s = 20 \times 500/100 = 100$ V.
xxii.	**D**	Gamma rays have the shortest wavelength.
xxiii.	**B**	Sound is made of compressions and rarefactions.
xxiv.	**B**	An alpha particle is a helium nucleus.
xxv.	**A**	$A - 4 = 234$, $Z - 2 = 90$.

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Q1(b) Fill in the blanks (1 mark each)

i.	**moment (torque)**
ii.	**zero**
iii.	**lower**
iv.	**right-hand**
v.	**gamma radiation (rays)**

Q1(c) True / False (1 mark each)

i.	**FALSE**
ii.	**TRUE**
iii.	**TRUE**
iv.	**FALSE**
v.	**TRUE**

Q1(d) Matching (1 mark each)

i. → **b**	ii. → **e**	iii. → **a**	iv. → **d**	v. → **c**

Section B (10 marks per question)

Question 2

Part	Marks	Answer and working
a)	2	Clockwise = anticlockwise: 300 × 1.5 = 450 × d ⇒ d = 450/450 = **1.0 m** from the pivot on the other side (the girl sits nearer the pivot).
b)	1	(1) The net (resultant) force is zero. (2) The net moment about any point is zero (clockwise moments = anticlockwise moments).
c)	2	**Cone on its base:** stable equilibrium (C.G. rises when tilted, the cone returns). **Cone on its tip:** unstable equilibrium (C.G. falls when displaced, it moves further). **Cone on its side:** neutral equilibrium (C.G. stays at the same height, it stays in the new position).
d)	1	The heavy wide base keeps the centre of gravity low and gives a large base area, so the line of gravity stays inside the base even when the lamp is tilted.
e)	2	Moment 1 = 20 × 0.30 = 6 N m. Moment 2 = 15 × 0.50 = 7.5 N m. The **15 N force at 0.50 m** gives the greater moment.
f)	2	Two equal, opposite parallel forces act on the opposite sides of the tap handle; the net force is zero but they give a turning effect (moment = F × distance between the forces) — a couple.

Question 3

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Part	Marks	Answer and working
a)	2	(i) $v = g\,t = 10 \times 4 =$ **40 m/s**. (ii) $h = \tfrac{1}{2} g\,t^2 = 0.5 \times 10 \times 16 =$ **80 m**.
b)	2	In a vacuum only gravity acts, so both have the same acceleration g. In air the feather's large surface area gives a large air resistance compared with its weight, so it falls more slowly than the coin.
c)	2	$F = G\,m_1 m_2 / r^2 = 6.67 \times 10^{-11} \times 5 \times 8 / (0.4)^2 = 6.67 \times 10^{-11} \times 40 / 0.16 =$ **1.67×10^{-8} N**.
d)	1	Energy can neither be created nor destroyed; it can only be changed from one form to another (the total energy stays constant).
e)	2	Work = m g h = 60 × 10 × 3 = 1800 J. Power = W/t = 1800/6 = **300 W**.
f)	1	Use LED lamps / 5-star appliances, insulate the walls and roof, switch off unused appliances (any one).

Question 4

Part	Marks	Answer and working
a)	2	$F_2 = F_1 \times A_2 / A_1 = 50 \times 80/4 =$ **1000 N**. Principle: **Pascal's law** (pressure is transmitted equally through an enclosed liquid).
b)	1	For the same force the sharp edge has a very small area, so the pressure (P = F/A) is large.
c)	2	(i) $P = \rho\,g\,h = 1000 \times 10 \times 2 =$ **20 000 Pa**. (ii) $F = P \times A = 20\,000 \times (2 \times 1.5) =$ **60 000 N**.
d)	2	$Q = m\,c\,\Delta T = 0.5 \times 4200 \times 50 =$ **105 000 J**.
e)	2	Heat lost = heat gained: $1 \times c \times (90 - T) = 2 \times c \times (T - 15) \Rightarrow 90 - T = 2T - 30 \Rightarrow 3T = 120 \Rightarrow$ **T = 40 °C**.
f)	1	Water has a high specific heat capacity, so it absorbs a large amount of heat from the engine for a small rise in its temperature.

Question 5

Part	Marks	Answer and working
a)	2	**Wire X** obeys Ohm's law (straight line through the origin). From the graph, V = 8 V when I = 2 A, so $R = V/I = 8/2 =$ **4 Ω**.
b)	2	(i) $I = P/V = 60/240 =$ **0.25 A**. (ii) $R = V/I = 240/0.25 =$ **960 Ω**.
c)	1	The filament has a much higher resistance (thin tungsten wire), so more heat is produced in it ($H = I^2Rt$).
d)	2	$H = I^2Rt = 3^2 \times 20 \times 5 = 9 \times 100 =$ **900 J**.
e)	1	Length, cross-sectional area, material, temperature (any two).
f)	2	For a given power P = V I, a higher voltage gives a smaller current. The heat loss I^2R in the transmission lines is then much smaller, so less energy is wasted.

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Question 6

Part	Marks	Answer and working
a)	2	Using **Fleming's left-hand rule**: first finger = field (to the right), second finger = current (out of the page); the thumb points **upwards (towards the top of the page)**, so the force on the wire is upward.
b)	1	Rotate the coil faster, or use a stronger magnet, or use more turns on the coil, or a larger coil (any one).
c)	3	(i) $V_s = V_p \times N_s/N_p = 240 \times 100/400 =$ **60 V**. (ii) $V_pI_p = V_sI_s \Rightarrow I_s = 240 \times 0.5 / 60 =$ **2 A**. (It is a step-down transformer.)
d)	1	A DC gives a steady magnetic flux; there is no change of flux in the core, so no emf is induced in the secondary coil.
e)	2	AC generator: **slip rings**; output **alternates** in direction. DC generator: **split-ring commutator**; output flows in **one direction**.
f)	1	Lamination reduces eddy currents, so less energy is wasted as heat in the core.

Question 7

Part	Marks	Answer and working
a)	2	$\lambda = c/f = 3 \times 10^8 / (100 \times 10^6) =$ **3 m**.
b)	2	P to R is one complete **wavelength**; the distance from Q to the dotted line is the **amplitude**. The vibration is perpendicular to the direction of travel, so it is a **transverse** wave.
c)	2	It carries signals over a short range and does not pass through to other rooms or interfere with other devices. Limitation: it travels in straight lines, so the path must not be blocked (line of sight).
d)	1	Skin burns / skin cancer / eye damage (any one).
e)	1	Digital signals are less affected by noise and can be regenerated without loss (any one: easier to store, process, or secure).
f)	2	Radio waves have a long wavelength, comparable to the hill, so they diffract around it (and can be reflected by the ionosphere). Light has a very short wavelength and does not diffract noticeably around the hill.

Question 8

Part	Marks	Answer and working
a)	1	X = $\mathbf{^{222}_{86}Rn}$ (A = 226 − 4 = 222; Z = 88 − 2 = 86).
b)	2	$^{214}_{82}Pb \rightarrow$ $\mathbf{^{214}_{83}Bi + {}^{0}_{-1}e}$ (mass number unchanged, atomic number increases by 1).
c)	2	(i) **Alpha particles**. (ii) **Gamma rays**.
d)	1	Use long tongs and keep the source in a lead-lined box (or wear protective clothing / keep the exposure time short).

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e)	1	Gamma rays are highly penetrating and kill bacteria even inside sealed packets, without heating the instruments.
f)	2	**Orbiter:** goes into orbit around a planet and studies it from above for a long time. **Rover:** lands and moves over the surface, collecting soil and rock samples.
g)	1	The wavelengths of light from distant galaxies are stretched, so the galaxies are moving away: the universe is expanding.

Question 9

Part	Marks	Answer and working
a)	2	Thick insulated walls and roof; double-glazed windows; large south-facing windows for sunlight; draught-proof doors; solar water heater (any two).
b)	2	KE = ½ m v^2 = 0.5 × 1000 × 400 = **200 000 J**. Work done by the brakes = change in KE = **200 000 J**.
c)	2	Yes. (1) Hydropower is low in winter, solar is strong then, so it improves energy security. (2) Solar is clean and renewable and reduces imports.
d)	2	F = 40 × 10 = 400 N; P = F/A = 400/0.04 = **10 000 Pa**.
e)	1	It has a high centre of gravity and a narrow base, so it can easily topple; fixing it to the wall prevents the line of gravity leaving the base.
f)	1	Advantage: needs little fuel and can visit several planets. Limitation: collects data only for a short time (it does not orbit or land).

Marking guide: award marks for the correct formula, substitution and final answer with unit (for calculations: 1 mark for method, 1 mark for the answer with unit in 2-mark parts). Accept g = 9.8 m/s^2 with correspondingly adjusted answers.

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